Michael Cusack-Nelkin

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Education

Columbia University | M.A. in Mathematics | New York, NY | Aug 2025 - Dec 2026

- GPA: 3.6 / 4.00
- GRE Score: 169V - 166Q - 4.5W

The College of William & Mary | B.S. in Physics | Williamsburg, VA | Aug 2018 - May 2022

- GPA: 3.7 / 4.00
- Minors in Data Science and Mathematics
- Honors: Dean's List, *magna cum laude*

Work Experience

Deloitte Consulting | Strategy and Analytics Analyst - State of Texas | Austin, TX | Oct 2022 - Jun 2025

- Led four developers and testers in delivering a fraud recovery system for the Texas Office of Inspector General, identifying and reovering $10 million in previously unrecoverable fraudulent Medicaid claims.
- Presented technical progress, design decisions, and testing results to a mixed audience of technical and non-technical clients multiple times per week, maintaining stakeholder support throughout development, implementation, and claim recovery.

DLA Piper | Financial Analyst Intern | New York, NY | Jun 2020 - Sep 2020

- Analyzed practice-level profitability and billing data and prepared weekly reports for finance leadership. Allowed for finance leaders to better understand the profitablility and performance of individual teams and clients.

Gilder, Gagnon, Howe & Co. | Research Analyst Intern | New York, NY | Mar 2018 - Jun 2018

- Analyzed public online gaming companies' business models, competitive position, and financials using 10-Ks, 10-Qs, and earnings calls and contributed to a quarterly research summary.
- Participated in investment committee meetings and portfolio company interviews in agrotech, biotech, and live entertainment.

Research and Competitions

Millennium x Columbia University Trading Competition | New York, NY

- Led a team of four to win a Millennium Management ETF arbitrage competition for Columbia students. Used a rolling z-score to estimate a hedging relationship between the ETF and its underlying components, and traded a synthetic basket of the underlyings and ETF.

High Sharpe, Low Capacity: Optimally Trading the Nasdaq Closing Auction Imbalance | Summer 2026

- Built a statistical arbitrage system combining a return forecast, 10-factor covariance model, spread costs and price impact estimates into a constrained portfolio optimization problem in Python using CVXPY.
- Backtested the strategy over 110 days and 100 stocks. Estimated an 8.53 Sharpe ratio at a maximum capacity of $91.3 million GMV.
- Identified fragility in capacity and latency requirements that favor use within a larger portfolio's execution process.

Adaptive Execution Strategies Lead to Robust Improvements Over Naive Benchmark | Spring 2026

- Co-developed an execution simulator using L2 Nasdaq order book data, supporting market, resting limit, and IOC orders under configurable latency and stochastic replenishment assumptions.
- Designed an execution strategy that used book imbalance to adjust fill aggression and reduced simulated arrival costs by 55%-89% versus a naive benchmark across three studied assets.

Technical Skills

- Finance: financial statement analysis, strategic analysis, SEC filings, research writing, market structure, competitive analysis.
- Programming and Methods: Python, NumPy, Pandas, Polars, scikit-learn, statsmodels, XGBoost, CVXPY, Git, portfolio optimization, factor risk modeling, market microstructure analysis, data cleaning and normalization.